

AMINA BAYOUDH

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 Ariana Soghra, Tunisie
 [Portfolio](#)

ABOUT ME

I'm a second-year IT engineering student (4th year at ESPRIT) with a strong passion for coding, software development, and building secure, reliable systems. My studies focus on network infrastructure and data security, and I'm equally driven by my enthusiasm for artificial intelligence. I enjoy exploring how intelligent systems can enhance performance, strengthen security, and solve real-world problems.

EDUCATION

- 2023 - Present:
ESPRIT (Private Higher School of Engineering and Technology)
- 2019 - 2022:
Abu Kassem Chebbi Secondary School - Baccalaureate

ACADEMIC PROJECTS

ENIGMA Next-Generation SOC for the Banking Sector November 2025 - May 2026

Description:
Fully automated open-source SOC for banking, real-time threat detection, AI-assisted triage, automated endpoint response, and complete IAM across five attack scenarios with zero manual intervention.

- Key Features:**
- Network segmentation (WAN, DMZ, LAN, SOC) via pfSense/OPNsense; centralized SIEM/XDR with Wazuh
 - SOAR automation via Shuffle covering DDoS, brute-force, malware, privilege escalation, and phishing
 - AI-driven DDoS detection using a Transformer model on live TShark captures
 - LLM-based alert analysis for false-positive reduction and auto-remediation
 - Multi-layer phishing pipeline: CrewAI + XGBoost + PhishTank with automatic Gmail quarantine
 - Full IAM: Active Directory + MidPoint IGA (JML lifecycle) + Keycloak SSO

LotusAura – Medical Centre Platform for Women & Children November - December 2025

AI-powered healthcare platform for Tunisian families covering women's health and children's developmental care. Centralizes family health management, medical education, and emotionally aware AI assistance.

- Highlights:**
- RAG-based AI with multilingual support (Darija, French, English) and speech-to-text/TTS
 - AI breast cancer awareness assistant with voice emotion recognition (Lotus Bloom)
 - Game-based child behavioral monitoring for early ADHD/autism indicators (AuraMind)
 - Secure cloud infrastructure with role-based access and privacy protection

Achievement: 4th Place — IEEE TSYP13 EMBS CIS WIE Challenge
Technologies: Next.js · NestJS · PostgreSQL · Prisma · FastAPI · Flask · LangChain · FAISS/Qdrant · NLP Transformers · OpenAI/Groq APIs

EZGO – Smart & Inclusive Urban Mobility App (Java, Symfony, PHP) January - May 2025

Inclusive smart transportation app (Java + Symfony) modernizing urban mobility in Tunisia with real-time tracking, ride-hailing, car rental, and strong accessibility features for users with disabilities.

- Key Features:**
- Role-based user management with face recognition and SMS-based 2FA
 - AI-powered accessibility tools: sign language recognition and virtual keyboard

CERTIFICATES

- [Fundamentals of Deep Learning](#)
- [Rapid Application Development with Large Language Models \(LLMs\)](#)
- [Generative AI with Diffusion Models](#)
- All other certificates are available in the following [link](#)

PROFESSIONAL EXPERIENCE

ESPRIT TECH - AI in Cybersecurity Intern

July-August 2025

Conducted applied research on network traffic analysis using deep learning for real-time DDoS attack detection.

- Designed and trained a Transformer-based model to classify DDoS vs. normal traffic patterns
- Deployed real-time pipeline with TShark for live traffic capture, normalization, and inference
- Added AI explainability layer to identify the most influential traffic features per detection
- Followed full CRISP-DM methodology from data understanding through deployment

CliniSys - Human and Social Training Intern

June 2023

Database management tasks with SQL; IT hardware procurement and HR operations exposure

SKILLS

- **Programming:** Python, Java, C, C++, PHP, JavaScript, HTML/CSS
 - **Frameworks:** Symfony, Spring Boot, .NET, React, Next.js, NestJS, FastAPI, Flask
 - **Databases:** SQL, PostgreSQL
 - **SOC & Monitoring:** Wazuh (SIEM/XDR), Shuffle (SOAR), Zabbix
 - **Threat Intelligence:** TheHive, Cortex, MISP
 - **IAM:** Active Directory, MidPoint IGA, Keycloak
 - **AI / ML:** Machine Learning, Deep Learning, Transformers, LLMs (RAG, Fine-tuning), NLP, CNN/LSTM, XGBoost, Anomaly Detection
 - **Networking:** pfSense, OPNsense, TShark, Docker
 - **Languages:** Arabic (Native), English (B2), French (B2)
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JOURNEY WITH IEEE

- IEEE ESPRIT SB Media Manager
- SIGHT Day Congress Program Manager & Speaker Coordinator - November 2025-May 2026
- Code2Cure 3.0 - SDC Technical Challenge, WIE Technical Representative - November 2025-May 2026
- IEEE IAS Annual Meeting of Tunisia 5.0 Media Manager - June 2025 - January 2026
- IEEE IES SYP Congress 2.0 Media & Communication Team member - February 2025 - September 2025
- IEEE WIE ESPRIT SBAG Chair - February 2025-January 2026
- IEEE WIE ESPRIT SBAG Media Manager - April 2024-April2025